

Homework 7.1

Problem 6

Derive the following two formulas for approximating derivatives and show that they are both $\mathcal{O}(h^4)$ by establishing their error terms.

Part 1

$$\begin{aligned}
 f'(x) &\approx \frac{1}{12h} [-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)] \\
 f(x+2h) &= f(x) + 2hf'(x) + 4h^2 \frac{f''(x)}{2} + 8h^3 \frac{f'''(x)}{6} + 16h^4 \frac{f^{(4)}(x)}{24} + 32h^5 \frac{f^{(5)}(\xi_1)}{120} \\
 f(x+h) &= f(x) + hf'(x) + h^2 \frac{f''(x)}{2} + h^3 \frac{f'''(x)}{6} + h^4 \frac{f^{(4)}(x)}{24} + h^5 \frac{f^{(5)}(\xi_2)}{120} \\
 f(x-h) &= f(x) - hf'(x) + h^2 \frac{f''(x)}{2} - h^3 \frac{f'''(x)}{6} + h^4 \frac{f^{(4)}(x)}{24} - h^5 \frac{f^{(5)}(\xi_3)}{120} \\
 f(x-2h) &= f(x) - 2hf'(x) + 4h^2 \frac{f''(x)}{2} - 8h^3 \frac{f'''(x)}{6} + 16h^4 \frac{f^{(4)}(x)}{24} - 32h^5 \frac{f^{(5)}(\xi_4)}{120} \\
 -f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h) \\
 &= f(x)[-1 + 8 - 8 + 1] + f'(x)h[-2 + 8 + 8 - 2] + f''(x)h^2 \left[-\frac{4}{2} + \frac{8}{2} - \frac{8}{2} + \frac{4}{2} \right] \\
 &\quad + f'''(x)h^3 \left[-\frac{8}{6} + \frac{8}{6} - \frac{8}{6} + \frac{8}{6} \right] + f^{(4)}(x)h^4 \left[-\frac{16}{24} + \frac{8}{24} - \frac{8}{24} + \frac{16}{24} \right] \\
 &\quad + \frac{h^5}{120} [-32f^{(5)}(\xi_1) + 8f^{(5)}(\xi_2) - 8f^{(5)}(\xi_3) + 32f^{(5)}(\xi_4)] \\
 &\Rightarrow -f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h) \\
 &= f'(x)[12h] + \frac{h^5}{120} [-32f^{(5)}(\xi_1) + 8f^{(5)}(\xi_2) - 8f^{(5)}(\xi_3) + 32f^{(5)}(\xi_4)] \\
 &\Rightarrow f'(x) = \frac{1}{12h} [-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)] \\
 &\quad - \frac{h^4}{1440} [-32f^{(5)}(\xi_1) + 8f^{(5)}(\xi_2) - 8f^{(5)}(\xi_3) + 32f^{(5)}(\xi_4)] \\
 &\therefore f'(x) = \frac{1}{12h} [-f(x+2h) + 8f(x+h) - 8f(x-h) + f(x-2h)] + \mathcal{O}(h^4)
 \end{aligned}$$

Part 2

$$\begin{aligned}
 f''(x) &\approx \frac{1}{12} h^2 [-f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h)] \\
 f(x+2h) &= f(x) + 2hf'(x) + 4h^2 \frac{f''(x)}{2} + 8h^3 \frac{f'''(x)}{6} + 16h^4 \frac{f^{(4)}(x)}{24} + 32h^5 \frac{f^{(5)}(x)}{120} \\
 &\quad + 64h^6 \frac{f^{(6)}(\xi_1)}{720}
 \end{aligned}$$

$$f(x+h) = f(x) + hf'(x) + h^2 \frac{f''(x)}{2} + h^3 \frac{f'''(x)}{6} + h^4 \frac{f^{(4)}(x)}{24} + h^5 \frac{f^{(5)}(x)}{120} + h^6 \frac{f^{(6)}(\xi_2)}{720}$$

$$f(x-h) = f(x) - hf'(x) + h^2 \frac{f''(x)}{2} - h^3 \frac{f'''(x)}{6} + h^4 \frac{f^{(4)}(x)}{24} - h^5 \frac{f^{(5)}(x)}{120} + h^6 \frac{f^{(6)}(\xi_3)}{720}$$

$$f(x-2h) = f(x) - 2hf'(x) + 4h^2 \frac{f''(x)}{2} - 8h^3 \frac{f'''(x)}{6} + 16h^4 \frac{f^{(4)}(x)}{24} - 32h^5 \frac{f^{(5)}(x)}{120} + 64h^6 \frac{f^{(6)}(\xi_4)}{720}$$

$$\begin{aligned} -f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h) \\ = f(x)[-1 + 16 - 30 + 16 - 1] + f'(x)h[-2 + 16 - 16 + 2] \\ + f''(x)h^2 \left[-\frac{4}{2} + \frac{16}{2} + \frac{16}{2} - \frac{4}{2} \right] + f'''(x)h^3 \left[-\frac{8}{6} + \frac{16}{6} - \frac{16}{6} + \frac{8}{6} \right] \\ + f^{(4)}(x)h^4 \left[-\frac{16}{24} + \frac{16}{24} + \frac{16}{24} - \frac{16}{24} \right] + f^{(5)}(x)h^5 \left[-\frac{32}{120} + \frac{16}{120} - \frac{16}{120} + \frac{32}{120} \right] \\ + \frac{h^6}{720} [-64f^{(6)}(\xi_1) + 16f^{(6)}(\xi_2) + 16f^{(6)}(\xi_3) - 64f^{(6)}(\xi_4)] \end{aligned}$$

$$\begin{aligned} \Rightarrow -f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h) \\ = f''(x)[12h^2] + \frac{h^6}{720} [-64f^{(6)}(\xi_1) + 16f^{(6)}(\xi_2) + 16f^{(6)}(\xi_3) - 64f^{(6)}(\xi_4)] \end{aligned}$$

$$\begin{aligned} \Rightarrow f''(x) &= \frac{1}{12h^2} [-f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h)] \\ &\quad - \frac{h^5}{12 * 720} [-64f^{(6)}(\xi_1) + 16f^{(6)}(\xi_2) + 16f^{(6)}(\xi_3) - 64f^{(6)}(\xi_4)] \end{aligned}$$

$$\therefore f''(x) = \frac{1}{12h^2} [-f(x+2h) + 16f(x+h) - 30f(x) + 16f(x-h) - f(x-2h)] + \mathcal{O}(h^4)$$

Problem 7

Part 1

$$f'''(x) \approx \frac{1}{h^3} [f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)]$$

$$\begin{aligned} f(x+3h) &= f(x) + 3hf'(x) + 9h^2 \frac{f''(x)}{2} + 27h^3 \frac{f'''(x)}{6} + 81h^4 \frac{f^{(4)}(x)}{24} + 243h^5 \frac{f^{(5)}(x)}{120} \\ &\quad + 729h^6 \frac{f^{(6)}(\xi_0)}{720} \end{aligned}$$

$$\begin{aligned} f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x) \\ = f(x)[1 - 3 + 3 - 1] + f'(x)h[3 - 6 + 3] + f''(x)h^2 \left[\frac{9}{2} - \frac{12}{2} + \frac{3}{2} \right] \\ + f'''(x)h^3 \left[\frac{27}{6} - \frac{24}{6} + \frac{3}{6} \right] + f^{(4)}(x)h^4 \left[\frac{81}{24} - \frac{48}{24} + \frac{3}{24} \right] + \dots \end{aligned}$$

Since the term $\frac{81}{24} - \frac{48}{24} + \frac{3}{24}$ does not cancel out for us, we can truncate our Taylor series here.

$$\begin{aligned}
 & f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x) \\
 &= f'''(x)h^3 \left[\frac{27}{6} - \frac{24}{6} + \frac{3}{6} \right] + \frac{h^4}{24} [81f^{(4)}(\xi_0) - 48f^{(4)}(\xi_1) + 3f^{(4)}(\xi_2)] \\
 &\Rightarrow f'''(x) = \frac{1}{h^3} [f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)] \\
 &\quad - \frac{h}{24} [81f^{(4)}(\xi_0) - 48f^{(4)}(\xi_1) + 3f^{(4)}(\xi_2)] \\
 &\therefore f'''(x) = \frac{1}{h^3} [f(x+3h) - 3f(x+2h) + 3f(x+h) - f(x)] + \mathcal{O}(h)
 \end{aligned}$$

Part 2

$$\begin{aligned}
 f'''(x) &\approx \frac{1}{2h^3} [f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h)] \\
 &= f(x)[1 - 2 + 2 - 1] + f'(x)h[2 - 2 - 2 + 2] + f''(x)h^2 \left[\frac{4}{2} - \frac{2}{2} + \frac{2}{2} - \frac{4}{2} \right] \\
 &\quad + f'''(x)h^3 \left[\frac{8}{6} - \frac{2}{6} - \frac{2}{6} + \frac{8}{6} \right] + f^{(4)}(x)h^4 \left[\frac{16}{24} - \frac{2}{24} + \frac{2}{24} - \frac{16}{24} \right] \\
 &\quad + f^{(5)}(x)h^5 \left[\frac{32}{120} - \frac{2}{120} - \frac{2}{120} + \frac{32}{120} \right]
 \end{aligned}$$

Since we have that $\frac{32}{120} - \frac{2}{120} - \frac{2}{120} + \frac{32}{120} \neq 0$, we can truncate our Taylor series here.

$$\begin{aligned}
 & f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h) \\
 &= f'''(x)h^3[2] + \frac{h^5}{120} [32f^{(5)}(\xi_1) - 2f^{(5)}(\xi_2) + 2f^{(5)}(\xi_3) - 32f^{(5)}(\xi_4)] \\
 &\Rightarrow f'''(x) = \frac{1}{2h^3} [f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h)] \\
 &\quad - \frac{h^5}{120} [32f^{(5)}(\xi_1) - 2f^{(5)}(\xi_2) + 2f^{(5)}(\xi_3) - 32f^{(5)}(\xi_4)] \\
 &\therefore f'''(x) = \frac{1}{2h^3} [f(x+2h) - 2f(x+h) + 2f(x-h) - f(x-2h)] + \mathcal{O}(h^2)
 \end{aligned}$$

Part 3

Since the second one is of order $\mathcal{O}(h^2)$, it is the better choice.

Problem 13

Suppose that $L = \lim_{h \rightarrow 0} f(h)$ and that $L - f(h) = c_6 h^6 + c_9 h^9 + \dots$. What combination of $f(h)$ and $f\left(\frac{h}{2}\right)$ should be the best estimate of L .

$$\begin{aligned}
 L - f(h) &= c_6 h^6 + c_9 h^9 + \dots \\
 L - f\left(\frac{h}{2}\right) &= c_6 \left(\frac{h}{2}\right)^6 + c_9 \left(\frac{h}{2}\right)^9 + \dots = \frac{1}{2^6} c_6 h^6 + \frac{1}{2^6} \frac{1}{2^3} c_9 h^9 + \dots
 \end{aligned}$$

$$\begin{aligned} \Rightarrow 2^6 \left(L - f\left(\frac{h}{2}\right) \right) &= c_6 h^6 + \frac{1}{2^3} c_9 h^9 + \dots \\ [L - f(h)] - \left[2^6 \left(L - f\left(\frac{h}{2}\right) \right) \right] &= c_9 \left(h^9 - \frac{h^9}{2^3} \right) + c_{12} \left(h^{12} - \frac{h^{12}}{2^6} \right) + \dots \\ \Rightarrow (2^6 - 1)L - f(h) - 64f\left(\frac{h}{2}\right) &= c_9 \left(h^9 - \frac{h^9}{2^3} \right) + c_{12} \left(h^{12} - \frac{h^{12}}{2^6} \right) + \dots \\ \Rightarrow (2^6 - 1)L &= f(h) + 64f\left(\frac{h}{2}\right) + c_9 \left(h^9 - \frac{h^9}{2^3} \right) + c_{12} \left(h^{12} - \frac{h^{12}}{2^6} \right) + \dots \\ \Rightarrow L &= \frac{1}{2^6 - 1} \left[f(h) + 64f\left(\frac{h}{2}\right) + c_9 \left(h^9 - \frac{h^9}{2^3} \right) + c_{12} \left(h^{12} - \frac{h^{12}}{2^6} \right) + \dots \right] \end{aligned}$$

Problem 14

Using Taylor series, derive the error term for the approximation

$$\begin{aligned} f'(x) &\approx \frac{1}{2h} [-3f(x) + 4f(x+h) - f(x+2h)] \\ -3f(x) + 4f(x+h) - f(x+2h) &= f(x)[-3 + 4 - 1] + f'(x)h[4 - 2] + f''(x)h^2 \left[\frac{4}{2} - \frac{4}{2} \right] + f'''(x)h^3 \left[\frac{4}{6} - \frac{8}{6} \right] + \dots \end{aligned}$$

Since $\frac{4}{6} - \frac{8}{6} \neq 0$, we can truncate the series here.

$$\begin{aligned} \Rightarrow -3f(x) + 4f(x+h) - f(x+2h) &= f'(x)[2h] + \frac{h^3}{6} [4f''(\xi_h) - 8f''(\xi_{2h})] \\ \Rightarrow f'(x) &= \frac{1}{2h} [-3f(x) + 4f(x+h) - f(x+2h)] - \frac{h^2}{12} [4f''(\xi_h) - 8f''(\xi_{2h})] \end{aligned}$$

Thus, the error term is $\frac{h^2}{12} [4f''(\xi_h) - 8f''(\xi_{2h})]$ which is order $\mathcal{O}(h^2)$.

Computer Problem 1

Program the algorithm in the text in which Richardson extrapolation is used repeatedly to estimate $f'(x)$. Test your program on the following:

$$\ln(x), x = 3$$

$$\frac{d}{dx} \ln(x) = \frac{1}{x} \Rightarrow f'(3) = \frac{1}{3}$$

The code is as follows:

```
function Richardson = R(f,x,h,M) % This program will take in as an input a
% function f, a point x, some step size h, and number of rows M and output
% the derivative of f at the point x to a certain accuracy.
```

```
p = @(x,h) (f(x+h)-f(x-h))/(2*h); % For the derivative, I will use the
% central difference, but any divided difference will work.

D = zeros(M); % This is the array that contains all the values of the table
% for Richardson extrapolation.

% Using the pseudocode from the book.

% This loop will initialize the first column of D.
for n = 0:M
    m = n+1; % Shift of index because MATLAB is weird.
    D(m,1) = p(x,h/(2^n));
end

% This loop will iterate through all the columns accordingly.
for k = 1:M % This is the loop that goes through the columns
    for n = k:M %This is the loop that goes through the rows.

        % Change of index.
        m = n+1;
        K = k+1;
        %Splitting the algorithm for clarity.
        DD = D(m, K-1)-D(m-1,K-1);
        DDD = DD/((4^K)-1);
        D(m,K) = D(m,K-1)+DDD; % Here is where we fill the rows.
    end
end

%The full matrix with all the inputs is here.
Richardson = D;
```

Choosing our step side to be .1 and our M to be 5, we get that:

```
>> f = @(x) log(x);
>> R(f,3,.1,5)

ans =

Columns 1 through 5

    0.333456872493362         0         0         0         0
    0.333364202675916    0.333358024688087         0         0         0
    0.333341049704230    0.333339506172784    0.333339212228097         0         0
    0.333335262365786    0.333334876543223    0.333334803057039    0.333334785766173         0
    0.333333815587675    0.333333719135801    0.333333700764255    0.333333696441538    0.333333695376705
    0.333333453896643    0.333333429783907    0.333333425191020    0.333333424110341    0.333333423844132

Column 6

         0
         0
         0
         0
         0
    0.333333423777824
```